Part I

1 Objectives of this section:

Goal for this section is to fit a model that simulates PTCH1's modulation of cholesterol accessibility properly. We hypothesize that PTCH1 works by reducing the level of accessible outer leaflet cholesterol. Model objectives when fitting are to replicate experimental changes in cholesterol accessibility and time needed for PTCH1 deactivation to take effect.

2 Reactions:

2.1 Initializations:

$[PTCH1] = 0.09$ nM

$[PTCH1_i] = 0$ nM

$[cholesterol_e] = 0.9 * C_{chol}$

$[cholesterol_e] = 0.1 * C_{chol}$

$[SHh] = 0$ nM

$[SMO_{cyt}] = 2.604$ nM

$[SMO_{ciliumcyt}] = 2.604$ nM

$[SMO_{cilial\ membrane}] = 0$ nM

$[SMO_{cilial\ membrane-chol}] = 0$ nM

2.2 Other values:

k_f: tuned

$k_r = 30 * k_f$: see literature values

$C_{chol} = 180000$ nM: see calculations

$C_{exchange}$: tuned (s^{-1})

$k_{cholesterol}$: tuned (s^{-1})

V_{max}: tuned (s^{-1})

K_M: tuned (nM)

k_{PTCH1}: tuned ($s^{-1} * nM^{-1}$)

$C_{cyt\ to\ membrane}$: tuned (s^{-1})

$C_{exchange\ SMO}$: tuned(s^{-1})

$k_{f\ embed}$: tuned (s^{-1})

$k_{r\ embed}$: $5s^{-1}$

C_{diff}: tuned

$k_{deg\ membrane}$

2.3 Reactions for Michaelis-Menten Model:

2.3.1 PTCH1 Portion

$\frac{d[PTCH1]}{dt} = k_f * [PTCH1] * [SHh] - k_r * [PTCH1_i]$

$\frac{d[PTCH1_i]}{dt} = \frac{-dPTCH1}{dt}$

$\frac{d[cholesterol_e]}{dt} = 0.9 * [cholesterol_i] - 0.1 * [cholesterol_e]$
$- k_{PTCH1} * [cholesterol_e] * [PTCH1]$

2.3.2 SMO Kinetics

$\frac{d[SMO_{cyt}]}{dt} = C_{diff} * ([SMO_{cilium\ cyt}] - [SMO_{cyt}])$

$\frac{d[SMO_{cilium\ cyt}]}{dt} = \frac{d[SMO_{cyt}]}{dt}$ - $k_{f\ embed} * [SMO_{cyt}] + k_{r\ embed} * [SMO_{cilial\ membrane}]$

$\frac{d[SMO_{cilial\ membrane}]}{dt} = k_{f\ embed} * [SMO_{cyt}]$ - $k_{r\ embed} * [SMO_{cilial\ membrane}] + k_{r\ SMO} * [SMO_{cilial\ membrane-chol}]$
- $k_{f\ SMO} * [SMO_{cilial\ membrane}] * [cholesterol_e]$

$\frac{d[SMO_{cilial\ membrane-chol}]}{dt} = k_{f\ SMO} * [SMO_{cilial\ membrane}] * [cholesterol_e]$ - $k_{r\ SMO} * [SMO_{cilial\ membrane-chol}]$

Signaling Rate $\alpha\ \lambda * [SMO_{cilial\ membrane}] + \gamma * [SMO_{cilial\ membrane-chol}]$ (to simplify, $\lambda = 0$, $\gamma = 0$; thus, only cholesterol-bound SMO contributes to signaling)

2.4 Reactions for Linear Model:

2.4.1 PTCH1 Portion

$\frac{d[PTCH1]}{dt} = k_f * [PTCH1] * [SHh] - k_r * [PTCH1_i]$

$\frac{d[PTCH1_i]}{dt} = \frac{-dPTCH1}{dt}$

$\frac{d[cholesterol_e]}{dt} = 0.9 * [cholesterol_i] - 0.1 * [cholesterol_e]$
$- k_{PTCH1} * [cholesterol_e] * [PTCH1]$

2.4.2 SMO Kinetics (same as above)

$\frac{d[SMO_{cyt}]}{dt} = C_{diff} * ([SMO_{cilium\ cyt}] - [SMO_{cyt}])$

$\frac{d[SMO_{cilium\ cyt}]}{dt} = \frac{d[SMO_{cyt}]}{dt}$ - $k_{f\ embed} * [SMO_{cyt}] + k_{r\ embed} * [SMO_{cilial\ membrane}]$

$\frac{d[SMO_{cilial\ membrane}]}{dt} = k_{f\ embed} * [SMO_{cyt}]$ - $k_{r\ embed} * [SMO_{cilial\ membrane}] + k_{r\ SMO} * [SMO_{cilial\ membrane-chol}]$
- $k_{f\ SMO} * [SMO_{cilial\ membrane}] * [cholesterol_e]$

$\frac{d[SMO_{cilial\ membrane-chol}]}{dt} = k_{f\ SMO} * [SMO_{cilial\ membrane}] * [cholesterol_e]$ - $k_{r\ SMO} * [SMO_{cilial\ membrane-chol}]$

Signaling Rate $\alpha\ \lambda * [SMO_{cilial\ membrane}] + \gamma * [SMO_{cilial\ membrane-chol}]$ (to simplify, $\lambda = 0$, $\gamma = 0$; thus, only cholesterol-bound SMO contributes to signaling)

3 Assumptions:

- Cholesterol distribution and concentration between leaflets in the cilium is assumed to be similar to those measured in synthetic vesicles
- Total PTCH1 levels remain constant (no degradation or translation is modeled for PTCH1 and only conversion between activated and inhibited is modeled.
- Calculating whole cell cholesterol concentration using molecule count per cell and estimated single cell volume (approximated with 13μM-radius sphere) results in good approximations of cholesterol-PTCH1 and cholesterol-SMO abundance ratios.
- A single scalar relates species abundance to their contribution to reaction rate in reactions involving aqueous ligands, so we will introduce a single tunable parameter, $c_{membrane_to_aq}$ to calculate rates involving membrane-bound reactants.
- Transport between the cytoplasm and cilium is described by the diffusion equation, where the exchange between compartments is proportional to the negative difference in concentration between compartments. We apply this for both intracellular exchange and lateral diffusion about the membrane. This can also modeled by first order equations.

4 Necessary calculations:

- Cell volume: $13\mu m$ radius sphere: $\frac{4}{3}\pi(13*10^{-6}m)^3*1000\frac{L}{m^3} = 9.2*10^{-12}L$
- Cilial volume: $\pi(0.2*10^{-6}m)^2 6.5*10^{-6}m*1000\frac{L}{m^3} = 8.2*10^{-16}L$
- $C_{cyt\ to\ cil}$ = Cell volume / cilial volume = 11000
- C: $\frac{10^9}{(N_A*Cellvolume)}*10^9\frac{M}{nM} = 180000nM$
- Observed Outer leaflet reduction in cholesterol (see paper): $M\beta CD$ cholesterol removal rate is proportional to cholesterol abundance in the outer leaflet.
- At $633\mu M$ $M\beta CD$, cholesterol removal proceeds with a half life of $30s \rightarrow 0.693/30s^{-1} = 0.0231s^{-1}$ without doxycyclone-induced PTCH1 expression. With PTCH1, cholesterol removal proceeds at $70s \rightarrow 0.693/70s^{-1} = 0.0099s^{-1}$. We pursue an objective 0.0099/0.0231 = 0.429X reduction of outer leaflet cholesterol by activated PTCH1. (https://pmc.ncbi.nlm.nih.gov/articles/PMC8654371/)
- Diffusion between compartments:

5 Literature Values

- Primary cilium length: $3\mu m - 10\mu m$ (middle of range used)
- Primary cilium radius: $0.2\mu m$ https://pmc.ncbi.nlm.nih.gov/articles/PMC11180893/
- Cholesterol molecules per cell: $10^9 https://www.ncbi.nlm.nih.gov/books/NBK26871/$
- Total protein molecules per cell: $10^9 https://pmc.ncbi.nlm.nih.gov/articles/PMC3910158/$

6 References